$$\begin{cases} E = E_1 + E_2 \\ E_1 = \dfrac{1}{2} E_0 \cos(2\pi x / \lambda_x - 2\pi z / \lambda_z - \omega t) \\ E_2 = \dfrac{1}{2} E_0 \cos(2\pi x / \lambda_x + 2\pi z / \lambda_z - \omega t - \pi) \end{cases}, \tag{4}$$

λ_0

$$\left(\frac{1}{\lambda_0}\right)^2 = \left(\frac{1}{\lambda_x}\right)^2 + \left(\frac{1}{\lambda_z}\right)^2 = \left(\frac{\omega}{2\pi c}\right)^2 . \tag{5}$$

m=1 $\lambda_z = 2d$,

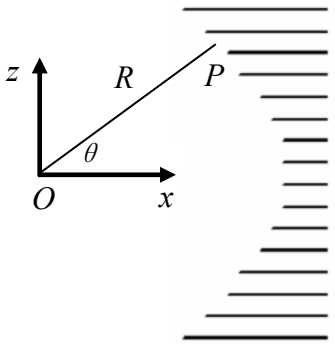
$$\lambda_x = \frac{1}{\sqrt{(2\pi c/\omega)^2 - (1/2d)^2}} . \tag{6}$$

3. x v_x

$$v_x = (\omega / 2\pi)\lambda_x = \frac{c}{\sqrt{1 - \left(\dfrac{\pi c}{\omega d}\right)^2}} . \tag{7}$$

P OP=R OP θ

$$L / c = R / c + (L - R\cos\theta) / v_x , \tag{8}$$



$$R = L(1 - \sqrt{1 - \left(\frac{\pi c}{\omega d}\right)^2})(1 - \sqrt{1 - \left(\frac{\pi c}{\omega d}\right)^2}\cos\theta) . \tag{9}$$